2. Finite State Automaton (FSA) — Strings Ending with ab

Aim: To implement a finite state automaton that accepts strings ending with ab, using only a and b.

CODE:

```
def fsa(s):
    state = 0

    for ch in s:
        if ch not in "ab":
            return False

        if state == 0:
            state = 1 if ch == 'a' else 0
        elif state == 1:
            state = 2 if ch == 'b' else 1
        else:
            state = 1 if ch == 'a' else 0

    return state == 2

# Test cases
tests = ["ab", "aab", "bab", "aba", "abb", "ba", "abc"]

for t in tests:
    print(t, "->", "Accepted" if fsa(t) else "Rejected")
```

main.py

Visualize

Run ▶

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# Test cases
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for t in tests:
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```

Output

```
ab -> Accepted
aab -> Accepted
bab -> Accepted
aba -> Rejected
abb -> Rejected
ba -> Rejected
abc -> Rejected
```